MYSQL DATABASE MANAGEMENT SYSTEM
LAB SHEET #3

Title:

- Perform set operations and various joins in Relational Database

Objectives:

- Import and use sample company database
- To understand and perform SQL set operations (Union, Intersect, Except/Minus)
- To practice different types of Joins (Inner, Left, Right, Full, Cross, Natural)

Background Concept:

- **UNION/UNION ALL**
  - SELECT <column_name> FROM <table1> UNION SELECT <column_name> FROM <table2>;
- **INTERSECT***
  - SELECT <column_name> FROM <table1> INTERSECT SELECT <column_name> FROM <table2>;
- **EXCEPT/MINUS***
  - SELECT <column_name> FROM <table1> EXCEPT SELECT <column_name> FROM <table2>;
- **OUTER JOIN***
  - SELECT <column_name> FROM <table_name1> <variable1> FULL OUTER JOIN <table_name2> <variable2> ON <variable1>.<column_name> = <variable2>.<column_name>;
- **INNER JOIN**
  - SELECT <column_name> FROM <table_name1> <variable1> INNER JOIN <table_name2> <variable2> ON <variable1>.<column_name> = <variable2>.<column_name>;
- **LEFT JOIN**
  - SELECT <column_name> FROM <table_name1> <variable1> LEFT JOIN <table_name2> <variable2> ON <variable1>.<column_name> = <variable2>.<column_name>;
- **RIGHT JOIN**
  - SELECT <column_name> FROM <table_name1> <variable1> RIGHT JOIN <table_name2> <variable2> ON <variable1>.<column_name> = <variable2>.<column_name>;
- **CROSS JOIN**
  - SELECT <column_name> FROM <table_name1> <variable1> CROSS JOIN <table_name2> <variable2>;
- **NATURAL JOIN**
  - SELECT <column_name> FROM <table_name1> NATURAL JOIN <table_name2>;

Lab works:

1. List all id, name and salary of both managers and employees in single table with duplicating the details of same employee
2. List the names of employees including managers together.
3. List all female employees and male managers together
4. List the various categories of salary in the company
5. List CROSS JOIN result of employees and departments
6. Display managers and departments using NATURAL JOIN.
7. Display names of employee and department names.
8. List employees along with their manager's name
9. Display all employees and their department names (include employees without departments)
10. List all employees with their manager's name (include employees without manager)
11. Display all departments and managers (include departments without managers)
12. List out the ids of managers who are assigned to employees
13. Display name of employees without managers
14. Show department code, department name, manager name, employee name, employee designation the above schema